Cloud LearnX

Mathematics

Transformations and Similar shapes — Practice Worksheet

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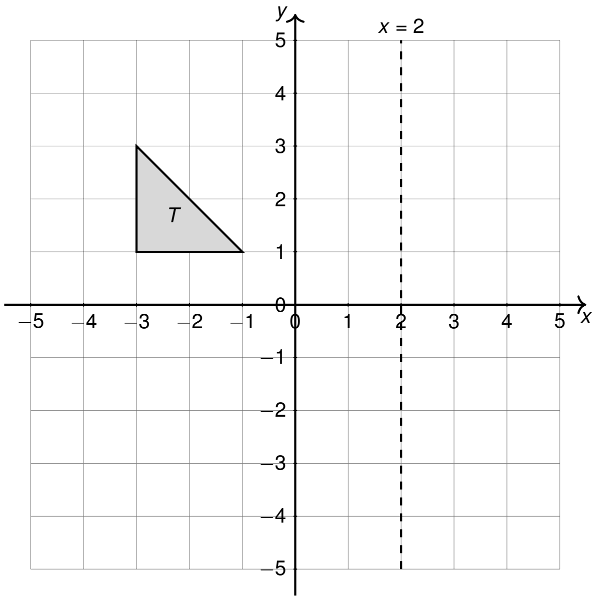
Topic: Transformations and Similar shapes

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

1 Shape T is drawn on the grid below. The line with equation $x = 2$ is also shown.

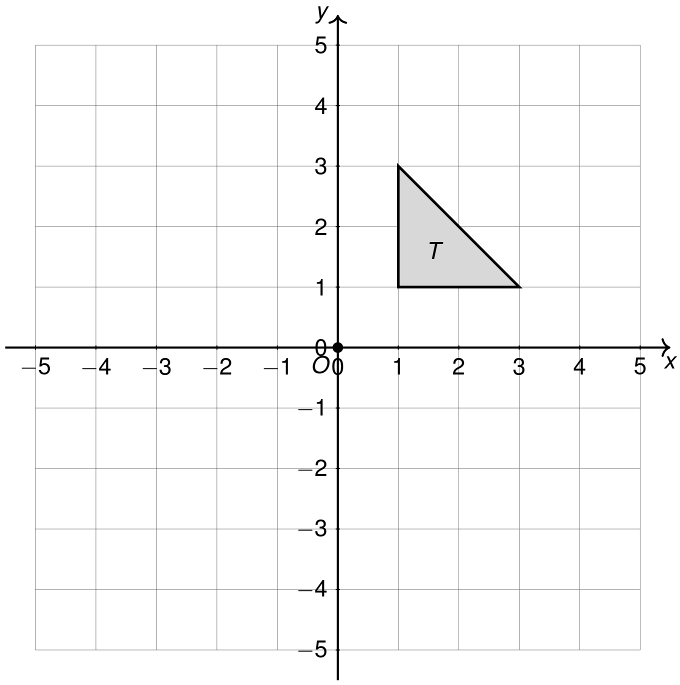


Reflect shape T in the line $x = 2$.
Label the image T'.

(2)

(Total for Question 1 is 2 marks)

2 A mission planner is mapping a triangular solar panel, T, on a satellite diagram using a coordinate grid, with the centre of the satellite's rotating arm marked at the origin $O(0, 0)$.



Rotate triangle T 90° clockwise about the centre $O(0, 0)$.

Draw and label the image T' on the grid above.

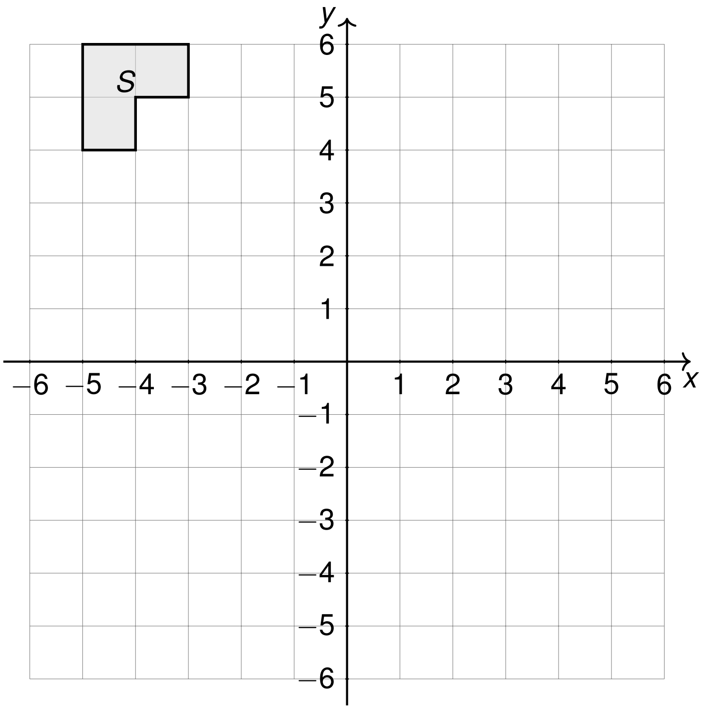
....

(2)

(Total for Question 2 is 2 marks)

3 A biologist is examining a specimen under a microscope. The outline of a bacterium, shape S, is plotted on the grid below, which represents the squares etched onto the slide.

The biologist shifts the slide so that the specimen's image moves according to the column vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.



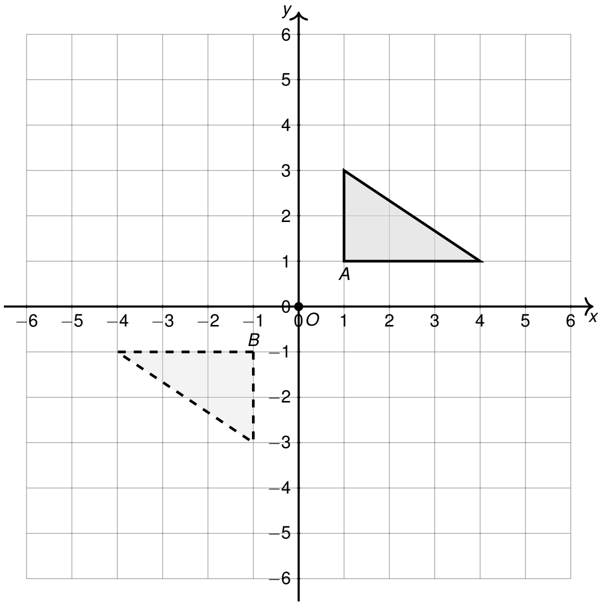
On the grid, draw the image of shape S after a translation by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

Label the image S'.

(3)

(Total for Question 3 is 3 marks)

4 Shape *A* and shape *B* are drawn on the grid below.

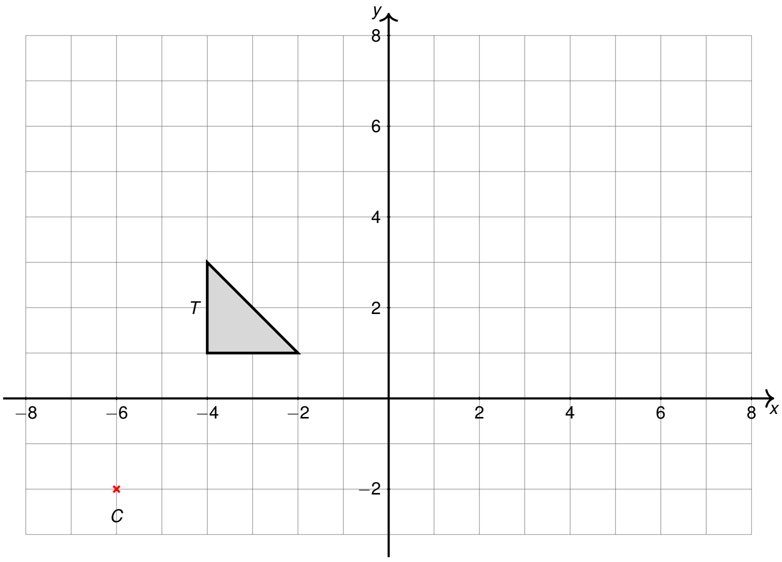


Describe fully the single transformation that maps shape *A* onto shape *B*.

(3)

(Total for Question 4 is 3 marks)

5 An astronomer is sketching a distant nebula as seen through a telescope. The lens produces a magnified image of shape T shown on the grid below, enlarged from a centre of enlargement C.



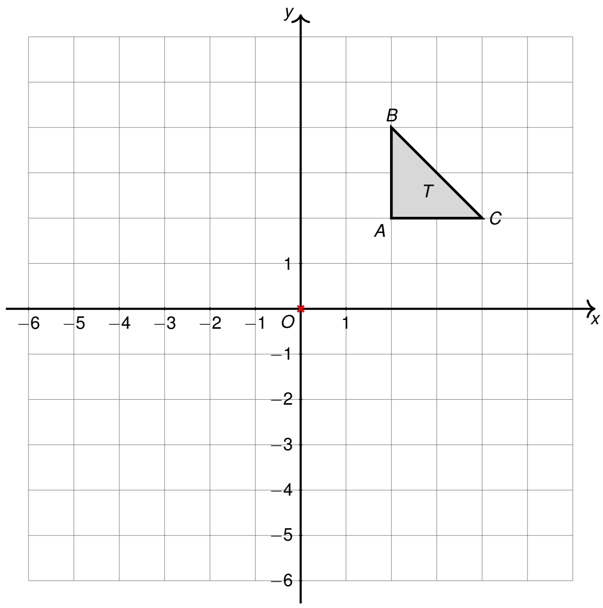
Enlarge shape T by scale factor 2, using C as the centre of enlargement.

....

(3)

(Total for Question 5 is 3 marks)

6 A space engineer is studying a triangular support bracket T on a control-panel schematic, drawn on a coordinate grid. She uses enlargements to model how the design changes when a scaled-down version is required for a satellite prototype (similar to how a microscope zooms out to shrink an image), and how a mirrored, inverted version is required for the opposite side of the panel.



Triangle T has vertices $A(2, 2)$, $B(2, 4)$, $C(4, 2)$.

(a) On a copy of the grid, enlarge triangle T by scale factor $\frac{1}{2}$, centre $(0, 0)$.
Label the image T_1.

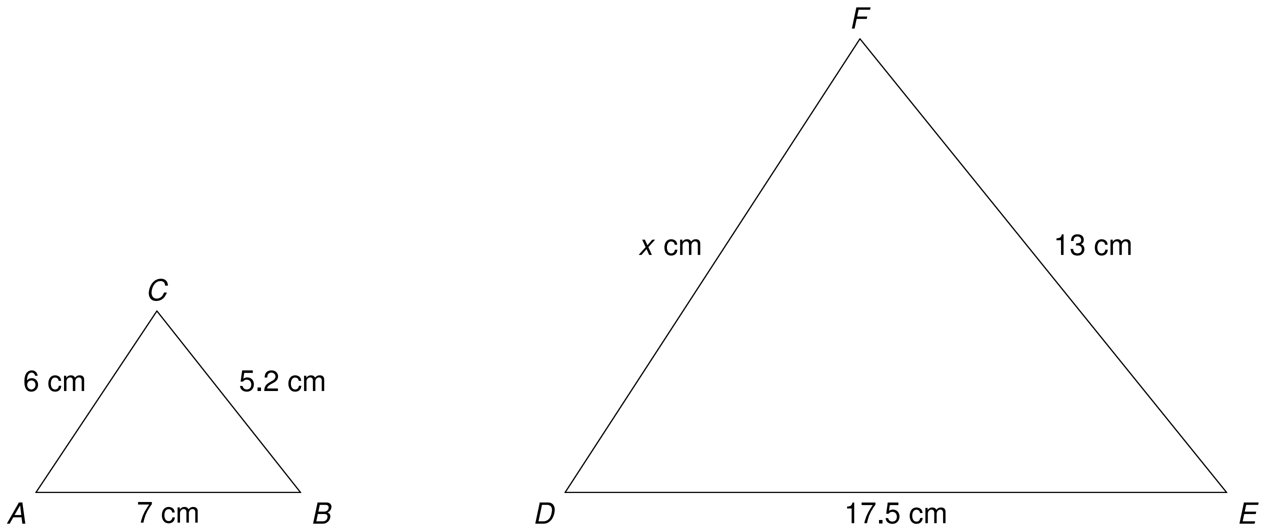
(2)

(b) On the same grid, enlarge the original triangle T by scale factor -1, centre $(0, 0)$.
Label the image T_2.

(2)

(Total for Question 6 is 4 marks)

7 A space agency uses two geometrically similar triangular support brackets to mount antenna dishes of different sizes onto a satellite frame. The smaller bracket is used on a test-model satellite, and the larger bracket is used on the full-size satellite. The two triangles *ABC* and *DEF* are similar, with *A* corresponding to *D*, *B* corresponding to *E*, and *C* corresponding to *F*.



Triangle *ABC* is similar to triangle *DEF*.

AB = 7 cm, *AC* = 6 cm, *BC* = 5.2 cm.

DE = 17.5 cm and *EF* = 13 cm.

Work out the length of *DF*.

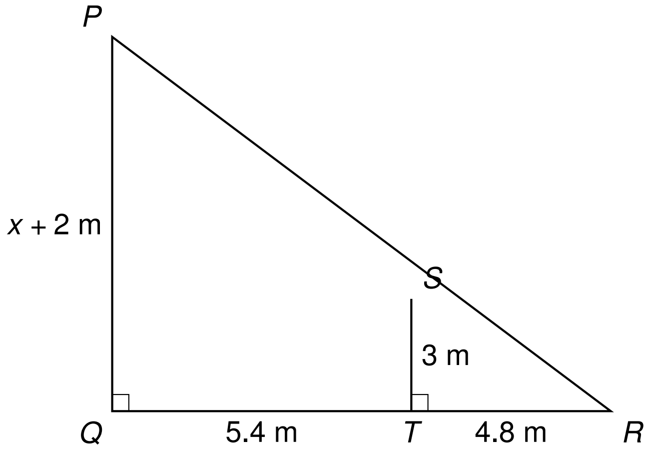
.... cm

(3)

(Total for Question 7 is 3 marks)

8 Astronomers estimating the height of a crater rim use a vertical measuring pole and the shadow it casts, compared with the shadow cast by the rim itself, to model the situation using similar triangles.

In the diagram, *PQ* represents the crater rim and *ST* represents the measuring pole. The Sun's rays are parallel, so *PR* and *SR* lie along the same straight line, and *QR* and *TR* lie along the same straight line.



$PQ = (x + 2)$ m is the height of the crater rim, $ST = 3$ m is the height of the pole.
$QT = 5.4$ m and $TR = 4.8$ m.

(a) Show that triangles *PQR* and *STR* are similar.

(2)

(b) Use similar triangles to find the value of x.

$x =$ ....

(2)

(Total for Question 8 is 4 marks)

TOTAL FOR PAPER IS ?? **MARKS**